

Article

Predicting the Concentration Levels of PM_{2.5} and O₃ for Highly Urbanized Areas Based on Machine Learning Models

Chao Wei ¹, Chen Zhao ^{2,*}, Yuanan Hu ^{3,*}  and Yutai Tian ²

¹ China National Environmental Monitoring Center, Beijing 100012, China; wcg523090@163.com

² State Key Laboratory of Environmental Criteria and Risk Assessment, Chinese Research Academy of Environmental Sciences, Beijing 100012, China; 18263765600@163.com

³ MOE Laboratory of Groundwater Circulation and Evolution, School of Water Resources and Environment, China University of Geosciences (Beijing), Beijing 100083, China

* Correspondence: zhaochen1030@sina.cn (C.Z.); yhu@cugb.edu.cn (Y.H.)

Abstract

The accurate real-time forecasting and impact factor identification of air pollutant levels are critical for effective pollution control and management. In this study, we implemented three machine learning algorithms, namely, Random Forest (RF), eXtreme Gradient Boosting (XGBoost), and Fully Connected Neural Network (FCNN), to predict PM_{2.5} and O₃ concentrations in the Beijing–Tianjin–Hebei region from 2019 to 2023. XGBoost outperformed the other algorithms and was further utilized to predict PM_{2.5} and O₃ concentrations and identify their controlling factors. The models could efficiently capture the spatial and temporal variations in the pollutants in the study area, and it was found that both anthropogenic sources and weather conditions can have significant impacts on air pollutant levels. PM₁₀ and CO were significantly correlated to PM_{2.5} levels, which could be attributed to their similar emission sources and dispersion characteristics in air. O₃ concentrations were greatly influenced by temperature and NO₂ due to their significant impacts on O₃ generation. This study demonstrates that XGBoost-based models are cost-effective tools for predicting PM_{2.5} and O₃ levels and identifying their controlling factors. These findings provide valuable insights for formulating effective air pollution prevention policies.

Keywords: air pollution; PM_{2.5}; O₃; machine learning; prediction



Academic Editors: Bin Ji and Giouli Mihalakakou

Received: 21 June 2025

Revised: 4 September 2025

Accepted: 9 October 2025

Published: 17 October 2025

Citation: Wei, C.; Zhao, C.; Hu, Y.; Tian, Y. Predicting the Concentration Levels of PM_{2.5} and O₃ for Highly Urbanized Areas Based on Machine Learning Models. *Sustainability* **2025**, *17*, 9211. <https://doi.org/10.3390/su17209211>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

With China's rapid industrialization and urbanization, air pollution has escalated, posing significant threats to human health and the ecological environment. PM_{2.5} and O₃ are globally recognized as critical outdoor air pollutants due to their detrimental health impacts [1]. In recent years, severe PM_{2.5} pollution events and their corresponding health risks sparked widespread public concern in China [2]. During the 13th Five-Year Plan (2016–2020), China implemented robust measures to reduce atmospheric PM_{2.5} levels, achieving a significant decline in average annual concentrations from 50 to 30 µg/m³ across 338 cities [3]. Despite these effective control efforts, regions such as central and northern China remain priorities for ongoing air quality improvement initiatives [4]. O₃, formed through photochemical reactions of nitrogen oxides and volatile organic compounds (VOCs) in the atmosphere, has garnered increasing attention from both the public and researchers. Exposure to high O₃ levels can seriously affect the respiratory and heart health of individuals and lead to negative outcomes for the natural environment [5].

In recent years, China's rapid industrialization and urbanization have driven remarkable economic growth but have also intensified severe air pollution challenges. Among the various air pollutants, PM_{2.5} and O₃ have emerged as key concerns, attracting significant attention from both the scientific community and the public [6].

PM_{2.5} originates from a complex and diverse range of sources. Industrial activities, including thermal power generation, iron and steel smelting, and cement manufacturing, are major contributors, emitting significant amounts of particulate matter containing heavy metals, sulfates, and nitrates [7]. In the transportation domain, fine particles from motor vehicle exhaust and road dust play crucial roles [8]. Additionally, coal combustion in residential settings, cooking fumes, and agricultural straw burning further exacerbate PM_{2.5} pollution levels [9].

Unlike directly emitted pollutants, O₃ formation is intricately tied to precursor compounds such as nitrogen oxides (NO_x) and volatile organic compounds (VOCs) [10]. These precursors, found in industrial emissions and vehicle exhausts, undergo complex photochemical reactions under sunlight and high-temperature conditions to produce O₃ [11]. Notably, during summer months with elevated temperatures, intensified photochemical activity often results in significantly higher O₃ concentrations [12].

Due to its minute particle size, PM_{2.5} can easily penetrate the respiratory tract, reaching the lungs and even entering the bloodstream via gas exchange [13]. This exposure can trigger numerous health issues, including respiratory inflammation and cardiovascular diseases, with prolonged exposure significantly increasing the risk of serious conditions such as lung cancer [14]. According to the World Health Organization, millions of people worldwide suffer premature deaths annually due to elevated PM_{2.5} concentrations. Similarly, O₃ poses substantial health risks. High O₃ levels irritate the respiratory mucosa, leading to symptoms such as coughing, wheezing, and breathing difficulties [15]. O₃ exposure can also damage lung cells and weaken the immune system, posing a particularly severe threat to vulnerable populations, including children and the elderly [16].

Despite China's significant efforts in air pollution prevention and control, which have led to notable reductions in PM_{2.5} concentrations in key regions, challenges persist. In industrially concentrated and densely populated areas, PM_{2.5} levels frequently exceed standards, and heavy pollution days continue to occur intermittently [17]. Concurrently, O₃ pollution has worsened, with an increasing number of days exceeding safe levels. O₃ has emerged as the primary pollutant impacting summer air quality, posing new challenges for air pollution prevention and control [18].

A thorough investigation of PM_{2.5} and O₃ pollution is crucial for developing targeted prevention and control strategies and protecting public health. Comprehensive understanding of their sources, health impacts, and current trends is vital for designing effective mitigation measures.

To devise effective pollutant management and control strategies, the accurate forecasting of the spatial and temporal distribution of pollutants under varying natural and anthropogenic conditions is essential. Chemistry transport models (CTMs) simulate the physical and chemical processes driving the formation and dispersion of atmospheric contaminants [19]. However, the accuracy of CTM outputs heavily relies on the quality of input data, such as emission inventories, and model parameters. Additionally, their application is often constrained by the high computational costs associated with large-scale numerical simulations. In contrast to CTMs, statistical techniques adopt a data-driven approach, predicting pollutant concentrations by uncovering patterns, trends, and relationships within observational data [20]. Traditional multivariate methods, such as correlation matrices and Principal Component Analysis (PCA), are widely employed to assess the degree of correlation among variables and identify potential sources of air pollutants [21,22]. Land-use

regression models and Geographically Weighted Regression (GWR) have been employed to evaluate spatial variations and identify the causes of outdoor air pollution [23,24]. However, these methods struggle to capture nonlinear behaviors and complex interactions between air pollutants and covariates, particularly in large-scale spatiotemporal studies. Additionally, they face challenges in accurately predicting pollutant concentrations using historical data.

Accurate prediction of PM_{2.5} concentrations is critical for effective air quality monitoring and mitigating associated health risks. Machine learning approaches have gained prominence in modeling the complex spatial and temporal variability of PM_{2.5}, owing to their ability to capture nonlinear relationships and integrate diverse data sources. Fully Connected Neural Networks (FCNNs), often combined with deep learning architectures like Long Short-Term Memory (LSTM) networks, have demonstrated strong potential for fine-grained urban air quality forecasting. For example, Han et al. (2021) developed the Deep-AIR framework, which integrates Convolutional Neural Networks (CNNs) and LSTM to effectively model spatial interactions and temporal dependencies, achieving high accuracy in predicting PM_{2.5} concentrations in metropolitan areas [25].

Random Forest (RF) models are widely favored for their robustness and interpretability. Hu et al. (2017) employed a Random Forest algorithm, integrating aerosol optical depth (AOD), meteorological data, and land-use variables, to estimate daily PM_{2.5} concentrations across the conterminous United States [26]. Their model achieved strong predictive performance, with an R^2 of 0.80 in cross-validation, demonstrating reliable spatial estimation of PM_{2.5} levels. Gradient-boosting methods, such as eXtreme Gradient Boosting (XGBoost), have gained significant attention for their efficiency and high accuracy in handling complex, multi-source datasets, particularly in environmental prediction tasks. Zhang and Batterman (2013) [27] utilized XGBoost, incorporating a diverse set of spatiotemporal features, including meteorological variables (e.g., wind speed, temperature, and precipitation) and satellite-derived data (e.g., aerosol optical depth). Their results demonstrated that XGBoost significantly outperformed traditional regression models in predicting PM_{2.5} concentrations, achieving lower root mean square error (RMSE) and a higher coefficient of determination (R^2). Furthermore, their study underscored the importance of feature selection in enhancing model performance: by identifying and prioritizing the most influential spatiotemporal features, the model's interpretability and generalization capabilities were markedly improved, offering valuable insights for advancing air quality prediction frameworks [28].

In recent years, various machine learning algorithms, including CNNs [29], RF [30], XGBoost [31], FCNNs [32], and Extremely Randomized Trees (ERTs) [33], have been applied to forecast ground-level air pollutant concentrations. These methods have demonstrated superior performance compared with traditional statistical modeling techniques. Machine learning approaches offer robust nonlinear mapping capabilities for large, complex datasets, leveraging flexible model structures and efficient algorithms to capture intricate interactions effectively [34]. Notably, RF and XGBoost models have proven highly effective in forecasting SO₂, PM_{2.5}, and O₃ concentrations, as demonstrated by numerous studies reporting promising results [35]. For instance, Vu et al. utilized an RF technique to quantify trends in PM_{2.5} air quality driven by anthropogenic emissions in Beijing from 2013 to 2017, achieving results consistent with those from Chemistry Transport Models (CTMs) [36]. Similarly, FCNNs have been successfully applied to predict PM_{2.5} concentrations [37]. The performance differences among FCNNs, XGBoost, and RF arise from their distinct architectural characteristics and adaptability to data, influencing several key aspects. FCNNs, with their deep hierarchical structures, are prone to overfitting, particularly when training data is limited or noisy, as they tend to memorize complex patterns, including noise. In contrast,

XGBoost and RF, as ensemble methods, reduce this risk through built-in regularization (e.g., tree depth constraints in XGBoost) and bagging/boosting techniques that average predictions across multiple learners, improving generalization to unseen data. Regarding data compatibility, XGBoost and RF are well-suited for tabular data, effectively handling mixed feature types (e.g., continuous meteorological variables and categorical emission source data) with minimal preprocessing. Conversely, FCNNs often require extensive feature engineering, such as normalization or dimensionality reduction, and struggle to intuitively capture spatial or categorical relationships, as they process inputs as flattened numerical vectors. Regarding sensitivity to design and tuning, FCNNs are highly sensitive to network architecture (e.g., number of layers and neurons) and hyperparameters (e.g., learning rate and regularization strength), necessitating careful tuning to avoid underfitting or instability. XGBoost and RF, however, are more robust to parameter variations, delivering consistent performance across a wide range of settings, making them more practical for implementation. Computationally, FCNNs demand significant training time and resources (e.g., GPU acceleration) due to iterative backpropagation and large parameter spaces. In contrast, XGBoost and RF operate efficiently on standard hardware, with faster convergence and lower memory requirements, making them ideal for the large-scale environmental datasets typical in air quality research. These three models were chosen for this study to balance predictive accuracy and computational feasibility. While FCNNs excel at capturing non-linear spatiotemporal patterns, XGBoost and RF offer reliable performance with shorter training times, a crucial factor for real-world air quality forecasting and policy support applications [38,39].

This study developed and validated a suite of machine learning models—RF, XGBoost, and FCNNs—to predict $\text{PM}_{2.5}$ and O_3 concentrations in the Beijing–Tianjin–Hebei region. Model performance was thoroughly assessed using multiple error metrics (RMSE, MAE, and R^2) to determine the most effective algorithm, which was subsequently applied to forecast pollutant levels and identify key influencing factors for individual cities in the region. This study pursues two primary objectives: first, to improve the accuracy and reliability of air pollutant predictions by leveraging advanced algorithms, including deep learning and ensemble methods, which adeptly capture complex nonlinear relationships in environmental data; second, to identify the primary drivers of $\text{PM}_{2.5}$ and O_3 variations, such as meteorological conditions, emission sources, and topographical features. By achieving these aims, this research provides valuable insights into the practical application of data-driven models in air quality research. The findings are poised to inform evidence-based air pollution control policies, support targeted strategies for mitigating $\text{PM}_{2.5}$ and O_3 pollution, and establish a scientific foundation for protecting public health and advancing sustainable environmental management in the Beijing–Tianjin–Hebei region.

2. Materials and Methods

2.1. Study Area and Dataset

The Beijing–Tianjin–Hebei region (Figure 1) is located between $36^{\circ}5' \sim 42^{\circ}40'$ N latitude and $113^{\circ}27' \sim 119^{\circ}50'$ E longitude, including Beijing, Tianjin, and 11 cities in Hebei Province (Baoding, Tangshan, Langfang, Cangzhou, Qinhuangdao, Shijiazhuang, Zhangjiakou, Chengde, Handan, Xingtai, and Hengshui). As seen in Figure 1, Handan has a notably high $\text{PM}_{2.5}$ concentration, with the green bar for this city standing out significantly compared with others. Other cities like Xingtai, Shijiazhuang, and Hengshui also exhibit relatively high $\text{PM}_{2.5}$ concentrations. In contrast, Chengde has a comparatively low $\text{PM}_{2.5}$ concentration, with the green bar for this city being shorter in height than most others. Zhangjiakou also shows a relatively low $\text{PM}_{2.5}$ concentration. Chengde stands out for its relatively high O_3 concentration, with the light-blue bar representing this value being

quite prominent. Additionally, cities like Qinhuangdao and Tangshan also show relatively high O_3 concentrations. On the other hand, Handan has a low O_3 concentration, with its light-blue bar being notably short. Xingtai and Shijiazhuang also fall within the lower range of O_3 concentrations. In 2023, the total Gross Domestic Product (GDP) of the Beijing–Tianjin–Hebei region reached CNY 10.44 trillion, nearly doubling from CNY 5.53 trillion in 2013, accounting for 8.3% of the national GDP. The Beijing–Tianjin–Hebei region and its surrounding areas were once the regions with the highest emission amounts and intensity (emissions per unit area) of air pollutants in China. Although cities in the Beijing–Tianjin–Hebei region and surrounding areas account for less than 3% of the country’s land area, they emitted more than 15% of the country’s primary $PM_{2.5}$, 10% of sulfur dioxide and nitrogen oxides, and 8% of VOCs (volatile organic compounds) [40]. The emission intensity was 2 to 5 times the national average.

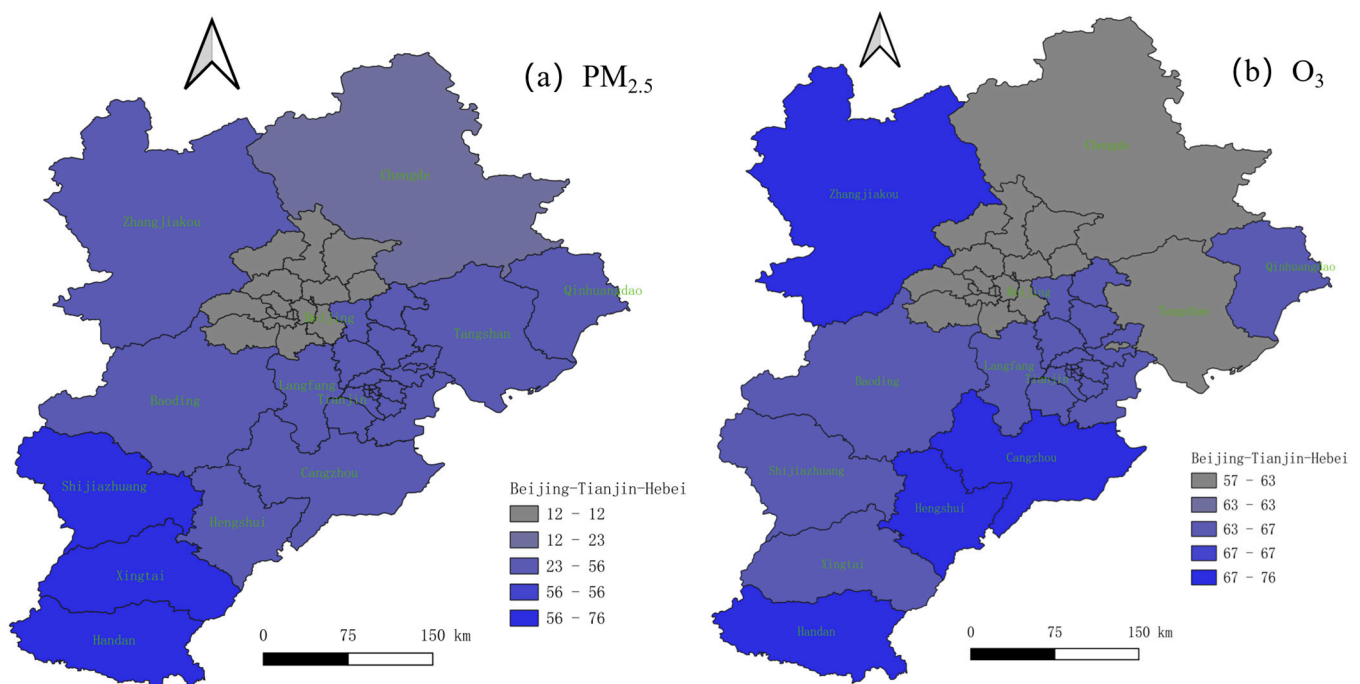


Figure 1. Spatial distribution of average hourly concentrations of $PM_{2.5}$ (a) and O_3 (b) in Beijing–Tianjin–Hebei region, China (2019–2023).

Meteorological variables such as temperature (T), surface pressure (P), relative humidity (RH), dew/frost point temperature (TDEW), wind speed (WS), wind direction (WD), precipitation (PREP), and all-sky surface shortwave downward irradiance (IRRA) are key factors in predicting $PM_{2.5}$ and O_3 concentrations because they directly and indirectly affect pollutant formation, dispersion, and removal.

Temperature influences atmospheric chemical reaction rates, accelerating photochemical reactions that increase O_3 production, and affects the emission of precursors for $PM_{2.5}$ formation [41]. Surface pressure impacts vertical mixing and atmospheric stability, thereby affecting pollutant dispersion [42]. Relative humidity and dew/frost point affect aerosol growth and secondary aerosol formation, influencing $PM_{2.5}$ levels [43]. Wind speed and direction control pollutant transport, affecting local concentrations [44]. Precipitation removes pollutants via wet deposition, reducing $PM_{2.5}$ [45]. Solar irradiance drives photochemical reactions essential for O_3 formation. Incorporating these meteorological factors improves the accuracy of $PM_{2.5}$ and O_3 prediction models.

The data used in this study are entirely sourced from the nationwide national air monitoring stations shared by the China National Environmental Monitoring Center. All

data have been reviewed, approved, and released by authoritative national agencies, ensuring their authenticity and reliability. The dataset spans the period from 2019 to 2023 and includes six air pollutants and eight meteorological factors at an hourly resolution for the Beijing–Tianjin–Hebei region. The concentrations of atmospheric pollutants, including PM_{2.5}, PM₁₀, CO, NO₂, SO₂, and O₃, were obtained from the National Environmental Monitoring Center of China (<https://air.cnemc.cn:18007>, accessed on 1 February 2024), while the meteorological information was obtained from the National Aeronautics and Space Administration (NASA, <https://power.larc.nasa.gov/data-access-viewer>, accessed on 1 February 2024) and includes temperature (T), surface pressure (P), relative humidity (RH), dew/frost point (TDEW), wind speed (WS), wind direction (WD), precipitation (PREP), and all-sky surface shortwave downward irradiance (IRRA). Table 1 provides the annotations, brief descriptions, and basic statistical information of the weather parameters. During the data preprocessing phase, missing values in the original dataset were filled using data from the previous time slot (i.e., using the forward-filling or last-observation-carried-forward method). Higher temperatures (T2M) accelerate O₃ formation by increasing the rates of NO₂ photolysis and VOC oxidation. Inversion layers (cold air trapped beneath warm air) at low T2MDEW can suppress vertical mixing, leading to PM_{2.5} accumulation [1]. Solar radiation (ALLSKY_SFC_SW_DWN) is critical for O₃ formation, as it drives NO₂ photolysis and initiates VOC oxidation cycles. For example, O₃ peaks in summer due to intense sunlight and high temperatures. Relative humidity (RH2M) affects aerosol hygroscopic growth: high humidity promotes water uptake by particles (e.g., sulfate or nitrate), increasing PM_{2.5} mass concentrations. Dew point (T2MDEW) indicates the likelihood of fog or dew, which can enhance aqueous-phase reactions (e.g., SO₂ oxidation) and reduce visibility. Regarding surface pressure (PS), high-pressure systems often bring clear skies and light winds, favoring O₃ formation but hindering PM_{2.5} dispersion. Wind (WS10M, WD10M) determines pollutant transport: strong winds (e.g., ≥5 m/s) enhance dispersion, while calm winds (≤2 m/s) trap pollutants near sources. Precipitation (PRECTOTCORR) is a primary removal mechanism for PM_{2.5}. Moderate-to-heavy rain (≥10 mm/day) efficiently scavenges particles, while light rain may have limited effect. SO₂ is a key precursor for sulfate aerosols. Its oxidation pathways include gas-phase reactions with hydroxyl radicals (OH) and aqueous-phase reactions in cloud droplets. Under sunlight, NO₂ photolyzes to NO and atomic oxygen (O), which reacts with O₂ to form O₃. CO is a byproduct of incomplete combustion. While it does not directly form PM_{2.5} or O₃, it consumes OH radicals (CO + OH → CO₂ + H), indirectly affecting O₃ production rates [46].

Table 1. Basic statistical information about meteorological factors.

Meteorological Factor	Description	Annual Mean (std)	Spring Mean (std)	Summer Mean (std)	Autumn Mean (std)	Winter Mean (std)
T	Temperature (°C)	12.29 (12.20)	13.20 (8.25)	25.71 (4.79)	12.64 (8.56)	−2.59 (5.22)
PS	Surface Pressure (kPa)	98.95 (0.996)	98.75 (0.75)	97.87 (0.46)	99.27 (0.71)	99.89 (0.65)
RH	Relative Humidity (%)	60.11 (20.95)	49.96 (21.58)	65.07 (21.10)	64.57 (19.29)	60.53 (18.27)
TDEW	Dew/Frost Point (°C)	3.52 (11.57)	1.06 (6.62)	17.33 (4.62)	5.25 (8.26)	−9.80 (4.87)
WS	Wind Speed (m/s)	3.52 (1.46)	4.23 (1.63)	3.07 (1.11)	3.24 (1.36)	3.53 (1.43)
WD	Wind Direction (Degrees)	189.74 (67.63)	193.21 (66.45)	172.03 (56.6)	192.64 (70.32)	197.69 (71.69)
PREP	Precipitation Corrected (mm/h)	0.082 (0.29)	0.041 (0.16)	0.22 (0.48)	0.061 (0.24)	0.010 (0.064)
IRRA	All-Sky Surface Shortwave Downward Irradiance (Wh/m ²)	174.25 (243.4)	219.59 (282.4)	221.13 (270.11)	145.70 (214.03)	110.46 (170.79)

2.2. Models Based on Machine Learning Algorithms

Modeling Framework and Variable Selection Principles: Theoretical and Empirical Insights for Lagged Variables in Time-Series Forecasting.

2.2.1. Fundamental Principles of the Modeling Framework and Variable Selection

Model I: Instantaneous Prediction Based on Contemporaneous Pollutant Concentrations

The core logic of Model I is grounded in the synergistic generation and mutual influences between pollutants within the same temporal framework. As a key atmospheric component, $\text{PM}_{2.5}$ exhibits intricate relationships with co-occurring pollutants. For instance, PM_{10} and $\text{PM}_{2.5}$ are inherently linked through shared emission sources (e.g., dust storms and coal combustion), with PM_{10} serving as a primary component and precursor to $\text{PM}_{2.5}$, thereby demonstrating a robust positive correlation [47]. O_3 acts as both a reactant and indicator in secondary aerosol formation. Photochemical reactions involving O_3 drive the conversion of SO_2 and NO_2 into sulfates and nitrates, establishing a nonlinear association where elevated O_3 concentrations often precede $\text{PM}_{2.5}$ surges, particularly under high humidity and low-wind conditions. SO_2 and NO_2 act as critical precursors for secondary $\text{PM}_{2.5}$ formation [48], with their conversion efficiency to particulate matter heavily dependent on meteorological parameters—aqueous-phase reactions under humid conditions, for example, accelerate sulfate formation [49]. CO, as a tracer for combustion sources, correlates with $\text{PM}_{2.5}$ emissions from traffic and industrial activities, providing complementary information on primary emission intensity [50]. This framework is tailored for short-term forecasting (0–12 h), harnessing real-time data to capture immediate pollution dynamics, such as abrupt emission peaks or rapid photochemical transformations.

Model II: Time-Series Prediction Incorporating Lagged Variables

Model II addresses the temporal lag in pollutant processes, where the previous day's environmental conditions and contaminant levels influence current-day concentrations. Chemical transformation lag is a defining characteristic, as the conversion of SO_2 and NO_2 to secondary aerosols requires hours to days. For instance, nitrate formation via nighttime N_2O_5 hydrolysis may not manifest in $\text{PM}_{2.5}$ until the following day [51]. Meteorological cumulative effects play a pivotal role. Stagnant conditions (e.g., low wind speed or temperature inversion) on day $(t-1)$ can trap pollutants, leading to elevated $\text{PM}_{2.5}$ on day t [52]. Emission inertia from industrial and transportation sectors ensures continuity in pollution sources, making day $(t-1)$ emissions a critical baseline for day t predictions.

Key variables in Model II include lagged $\text{PM}_{2.5}$ $(t-1)$, which represents pollutant accumulation under stable atmospheric conditions, serving as an initial condition for current-day modeling, and lagged precursors (PM_{10} , SO_2 , NO_2 , O_3 , and CO on day $(t-1)$, which capture the potential for secondary aerosol formation and regional transport. For example, elevated NO_2 on day $(t-1)$ signals increased nitrate aerosol production on day t [53].

2.2.2. Necessity and Advantages of Lagged Variables in Model II

- (1) Capturing long-range transport and cumulative effects: Lagged variables are indispensable for modeling cross-regional pollution transport. For instance, dust plumes or industrial emissions from upwind regions may take 6–18 h to reach the monitoring site, with day $(t-1)$ PM_{10} levels serving as a proxy for day t $\text{PM}_{2.5}$ impacts [54].
- (2) Enhancing the prediction stability: By incorporating historical data, Model II mitigates the impact of real-time data anomalies (e.g., sensor malfunctions) and captures seasonal emission patterns (e.g., winter coal heating cycles), thereby improving forecast robustness.

2.2.3. Introduction to Model Design

Two machine learning models, Models I and II, were developed to predict $\text{PM}_{2.5}$ and O_3 concentrations in the Beijing–Tianjin–Hebei region. Model I: This model employs three algorithms—RF, XGBoost, and FCNN—using contemporaneous pollutant datasets (e.g., SO_2, t at time t) and meteorological conditions (e.g., temperature T_t) as input variables.

The target outputs are pollutant concentrations at the same time (e.g., O_3, t). Model I is designed to identify potential sources and key factors influencing $PM_{2.5}$ and O_3 concentrations. Model II: Utilizing only XGBoost, this model incorporates historical pollutant data (e.g., O_3, t) and meteorological conditions at the predictive time ($t + 1$, i.e., one day after time t), such as temperature T_{t+1} , as inputs. The target outputs are pollutant concentrations at the predictive time (e.g., $O_3\{t + 1\}$). Model II enables forecasting of $PM_{2.5}$ and O_3 levels for the following day based on current pollution levels and forecasted weather conditions.

RF, XGBoost, and FCNNs are widely used machine learning algorithms. As a supervised learning algorithm for classification and regression tasks, RF constructs multiple decision trees using randomly selected subsets of data and features. The model enhances performance by aggregating the results of all trees. Based on gradient-boosted decision trees, XGBoost builds trees sequentially, with each tree trained on the residuals of its predecessor to improve predictive accuracy. As a fundamental neural network architecture, FCNN consists of multiple hidden layers with interconnected neurons. Each neuron in a layer is fully connected to all neurons in the preceding and following layers. Further details on these algorithms are provided in the Supporting Information. In this study, the GridSearchCV technique, as introduced by Wang et al. (2020) [55], was employed to optimize model configurations. This method involves two steps: grid search and k-fold cross-validation. Grid search identifies the optimal model parameters within a specified range based on the performance metrics from k-fold cross-validation. The dataset was split into training and testing sets in a 9:1 ratio using random sampling, and 10-fold cross-validation was conducted to build and assess model performance. The parameter set yielding the highest accuracy on the testing data was selected as the optimal configuration. Table S1 details the parameter settings for the models across the different machine learning algorithms.

2.3. Model Performance Evaluation

For evaluation of the model performance, multiple indicators were calculated using Equations (1)–(3) [56], namely, root mean square error (RMSE), mean absolute error (MAE), and R^2 .

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (1)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2} \quad (3)$$

where n is the number of samples in the test set, y is the predicted value, $\hat{y}$ is the true value, and $\bar{y}$ the mean value.

3. Results and Discussion

3.1. The Temporal and Spatial Distribution of $PM_{2.5}$ and O_3

Table 1 summarizes the basic statistical information of meteorological factors in the Beijing–Tianjin–Hebei region, revealing distinct seasonal variations. The annual average temperature is 12.29 °C, with hot summers (25.71 °C) and cold winters (−2.59 °C), while spring (13.20 °C) and autumn (12.64 °C) remain relatively mild. Relative humidity averages 60.11% annually, peaking in summer (65.07%) and reaching its lowest in spring (49.96%), likely due to reduced precipitation in spring (0.041 mm/h) compared with a significant increase in summer (0.22 mm/h). Wind speed is highest in spring (4.23 m/s) and lowest in summer (3.07 m/s), with wind direction varying significantly across seasons but predomi-

nantly from the south. Additionally, all-sky surface shortwave downward irradiance peaks in summer (221.13 Wh/m²) and drops to its lowest in winter (110.46 Wh/m²), reflecting clear seasonal variations in daylight hours.

Table 2 presents the results of the correlation analysis between pollutants and meteorological factors, highlighting the significant influence of weather conditions on pollutant concentrations in the Beijing–Tianjin–Hebei region. For example, temperature exhibits a negative correlation with PM_{2.5} levels, as colder weather often increases PM_{2.5} emissions from heating activities. Conversely, temperature is positively correlated with O₃ concentrations, as higher temperatures enhance O₃ production through photochemical reactions. Additionally, PM_{2.5} shows strong correlations with PM₁₀, CO, and NO₂, indicating that these pollutants likely originate from common sources, such as fuel combustion and industrial processes.

Table 2. Pearson’s correlation matrix of atmospheric pollutants and meteorological factors in the Beijing–Tianjin–Hebei region.

Item	PM _{2.5}	PM ₁₀	O ₃	SO ₂	NO ₂	CO	T	P	RH	TDEW	WS	WD	PREP	IRRA
PM _{2.5}	1.00													
PM ₁₀	0.74	1.00												
O ₃	−0.31	−0.19	1.00											
SO ₂	0.49	0.35	−0.21	1.00										
NO ₂	0.64	0.42	−0.65	0.59	1.00									
CO	0.83	0.47	−0.37	0.71	0.71	1.00								
T	−0.40	−0.27	0.69	−0.28	−0.53	−0.38	1.00							
PS	0.31	0.21	−0.56	0.22	0.42	0.30	−0.80	1.00						
RH	0.10	−0.04	−0.33	−0.14	0.25	0.10	−0.19	−0.05	1.00					
TDEW	−0.34	−0.28	0.49	−0.33	−0.38	−0.32	0.85	−0.78	0.33	1.00				
WS	−0.02	0.07	0.09	0.02	−0.13	−0.05	−0.04	0.02	−0.44	−0.28	1.00			
WD	0.05	0.08	−0.18	0.07	0.16	0.04	−0.19	0.04	−0.11	−0.25	0.09	1.00		
PREP	−0.12	−0.12	0.10	−0.14	−0.16	−0.08	0.21	−0.23	0.24	0.32	0.06	−0.17	1.00	
IRRA	−0.15	−0.08	0.34	0.06	−0.34	−0.08	0.44	−0.17	−0.68	0.05	0.26	−0.04	−0.04	1.00

Figure 2 illustrates the descriptive statistics of hourly average PM_{2.5} and O₃ concentrations in the Beijing–Tianjin–Hebei region from 2019 to 2023. PM_{2.5} concentrations exhibited a slight downward trend over the study period, accompanied by pronounced seasonal variations. Situated in the northern part of the North China Plain, the region experiences a typical warm temperate semi-humid and semi-arid continental monsoon climate, characterized by cold, dry winters and hot, rainy summers. As shown in Figure 2a, the annual mean PM_{2.5} concentration was highest in 2019 (49.39 µg/m³), gradually declining to 44.71 µg/m³ in 2020, 40.02 µg/m³ in 2021, and 37.4 µg/m³ in 2022, before a slight increase to 40.5 µg/m³ in 2023. Winters consistently recorded the highest mean PM_{2.5} concentrations across all years, particularly in 2019 (79.75 µg/m³), 2020 (79.79 µg/m³), and 2021 (52.42 µg/m³), driven by stable meteorological conditions, increased anthropogenic emissions, higher residential burning, lower temperatures, and reduced surface solar radiation [56]. In contrast, summer exhibited the lowest PM_{2.5} concentrations, with the minimum recorded in summer 2023 (20.69 µg/m³). Spring and autumn generally displayed moderate concentration levels.

Figure 2b shows that annual mean O₃ concentrations in the Beijing–Tianjin–Hebei region remained relatively stable from 2019 to 2023, with values of 66.86 µg/m³ in 2019, 65.9 µg/m³ in 2020, 63.45 µg/m³ in 2021, 69.19 µg/m³ in 2022, and 69.84 µg/m³ in 2023. Summer consistently exhibited the highest O₃ concentrations across all years, while winter recorded the lowest. Meteorological factors, such as ambient temperature and surface solar radiation, play a critical role in O₃ formation. Elevated O₃ levels in summer are driven by enhanced photochemical reactions due to high temperatures and strong solar radiation [57], whereas cold temperatures and weak solar radiation in winter suppress O₃ formation [58].

Spring and autumn generally showed moderate O_3 concentrations, with spring levels typically exceeding those in autumn in most years.

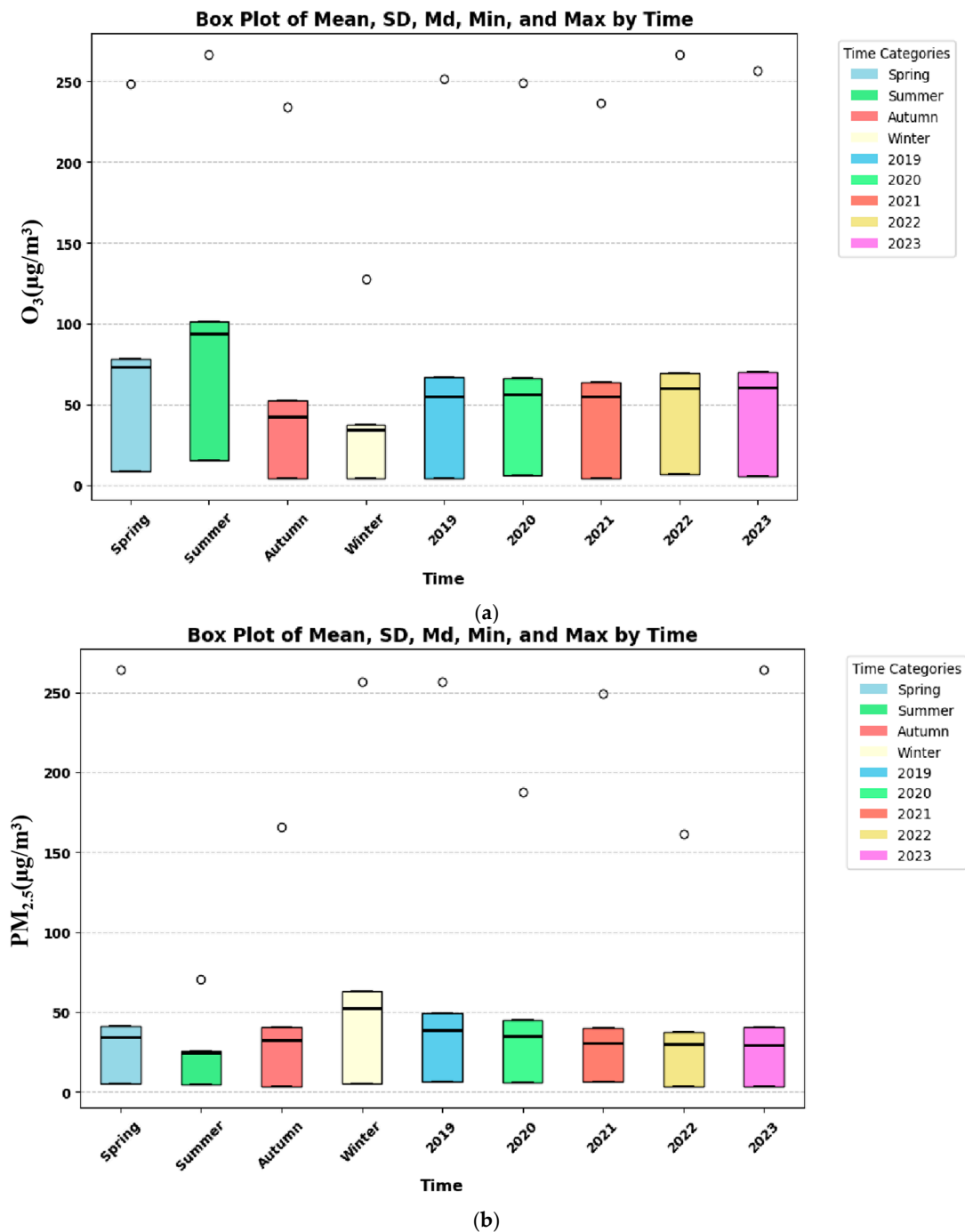


Figure 2. Boxplots of $PM_{2.5}$ (a) and O_3 (b) concentrations in Beijing–Tianjin–Hebei region, China (2019–2023).

Figure 3 illustrates the $PM_{2.5}$ and O_3 concentration levels across various cities in the Beijing–Tianjin–Hebei region. Notable spatial variations were observed in $PM_{2.5}$ concentrations, while O_3 levels showed relatively minor differences across cities. Among the cities, Handan, Xingtai, and Shijiazhuang exhibited higher $PM_{2.5}$ concentrations, whereas Chengde and Zhangjiakou recorded the lowest. For O_3 , Shijiazhuang, Hengshui, and Xingtai displayed relatively elevated levels.

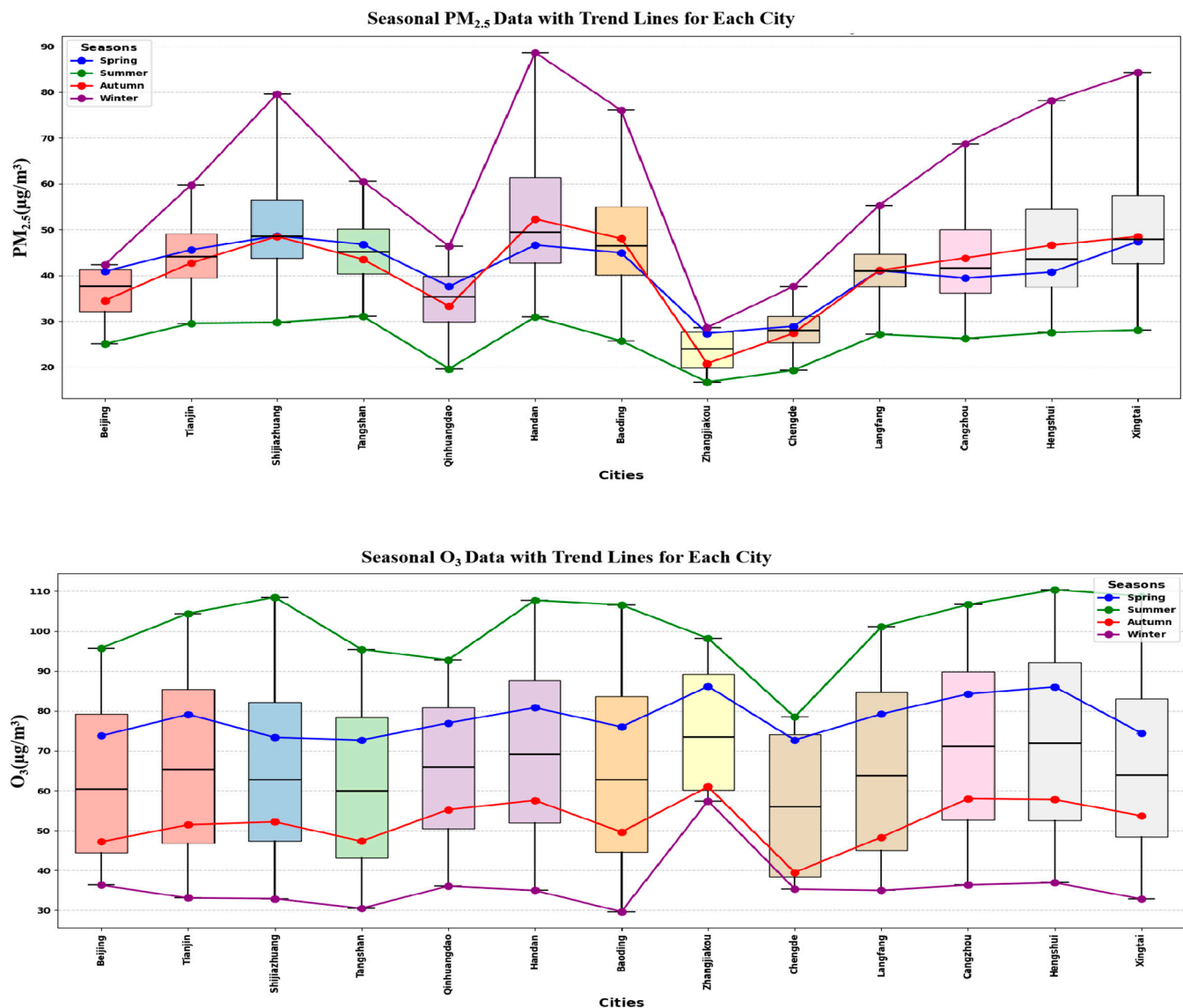


Figure 3. $\text{PM}_{2.5}$ and O_3 concentrations for different cities in the Beijing–Tianjin–Hebei region from 2019 to 2023.

3.2. $\text{PM}_{2.5}$ and O_3 Predictions Using Machine Learning Algorithms

In this study, two models employing three machine learning algorithms—XGBoost, FCNN, and RF—were developed to predict $\text{PM}_{2.5}$ and O_3 concentrations. Figures S1 and S2 present scatter plots of predicted versus observed values for the training and testing datasets, including fit lines and error estimations. The results indicate that the models performed well for most observations, with all three algorithms yielding comparable and highly effective predictions for $\text{PM}_{2.5}$ and O_3 . This underscores the potential of machine learning algorithms for applications such as identifying emission sources, detecting key influencing factors, and forecasting air pollution emergencies. Model performance was further assessed using statistical error metrics, summarized in Table 3 for both training and testing datasets. The models explained significant portions of the variance, with R^2 values ranging from 0.830 to 0.996, demonstrating their robust ability to capture underlying patterns in the data. However, FCNN exhibited notably higher prediction errors, as indicated by root mean square error (RMSE) and mean absolute error (MAE), compared with XGBoost and RF. Among the algorithms, XGBoost outperformed RF regarding R^2 , MAE, and RMSE, suggesting its superior predictive capability for $\text{PM}_{2.5}$ and O_3 concentrations.

in this study. Time-series plots of XGBoost-based model predictions alongside observed values are shown in Figure 4a,b, while mean concentrations for individual cities over the study period are depicted in Figure 4c,d. These results demonstrate that the models effectively captured both the temporal and spatial variations in pollutant concentrations.

Table 3. Error analysis of PM_{2.5} and O₃ predictions using three machine learning algorithms in the Beijing–Tianjin–Hebei region.

Model	Pollutant	Algorithm	RMSE		MAE		R ²	
			Training	Testing	Training	Testing	Training	Testing
Model I	PM _{2.5}	XGBoost	1.92	4.69	1.36	3.05	0.996	0.979
		RF	1.95	5.23	1.27	3.34	0.996	0.973
		FCNN	3.31	4.37	2.45	3.09	0.989	0.981
	O ₃	XGBoost	4.51	11.3	3.10	7.90	0.990	0.938
		RF	4.99	13.2	3.39	9.06	0.988	0.915
		FCNN	8.64	11.3	6.38	8.23	0.964	0.938
Model II	PM _{2.5}	XGBoost	5.88	13.0	4.04	8.60	0.966	0.830
		RF	16.1	17.5	11.15	12.2	0.745	0.693
		FCNN	11.0	13.4	8.16	9.68	0.881	0.821
	O ₃	XGBoost	7.01	15.8	4.87	11.2	0.976	0.878
		RF	15.1	17.4	11.4	12.9	0.891	0.853
		FCNN	13.7	15.4	10.2	11.3	0.910	0.885

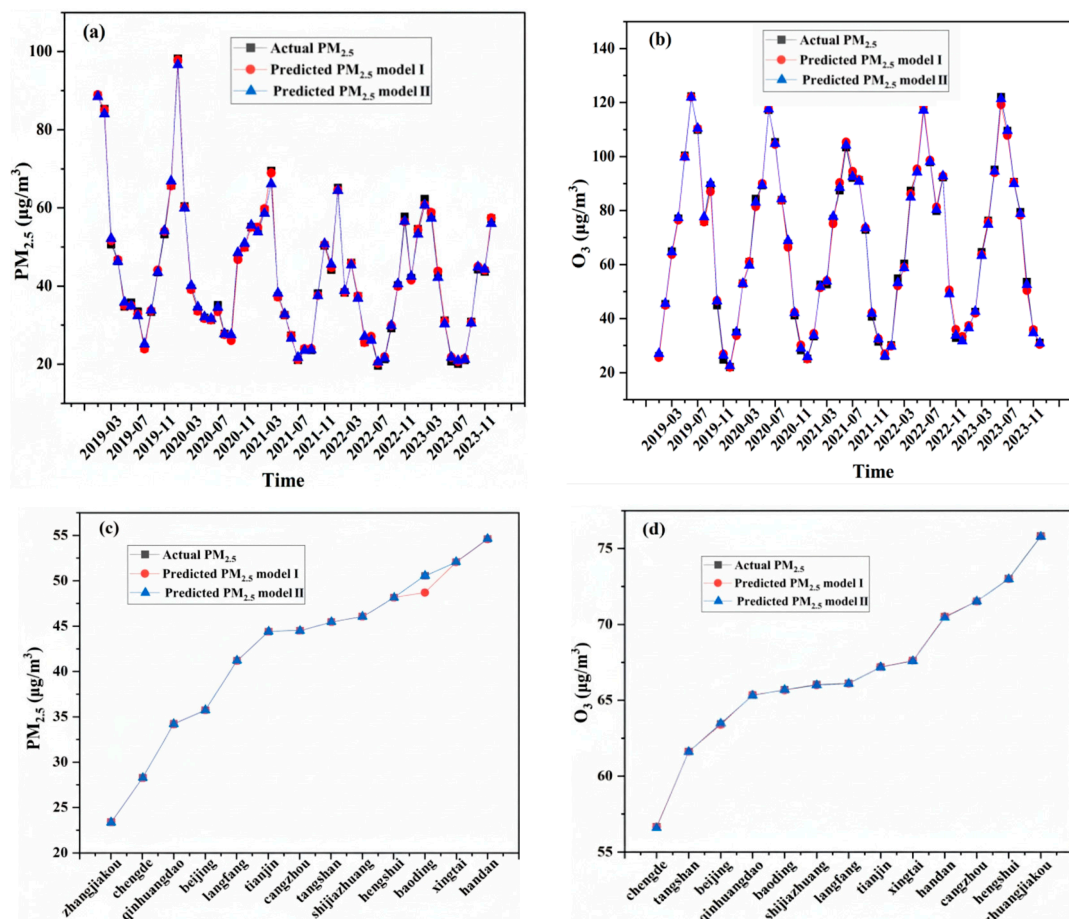


Figure 4. Concentrations of pollutants in the Beijing–Tianjin–Hebei region: (a) monthly PM_{2.5} concentrations; (b) monthly O₃ concentrations; (c) PM_{2.5} concentrations by city; (d) O₃ concentrations by city.

XGBoost was employed to predict PM_{2.5} and O₃ concentrations in the Beijing–Tianjin–Hebei region using Model I and Model II. As presented in Table 4, both models generally

produced reliable results across the investigated cities. For instance, Model I achieved R^2 values for $PM_{2.5}$ predictions on the testing dataset ranging from 0.914 (Langfang) to 0.976 (Shijiazhuang), and for O_3 predictions from 0.886 (Zhangjiakou) to 0.947 (Beijing). In contrast, Model II exhibited lower R^2 values overall, with $PM_{2.5}$ predictions on the testing dataset ranging from 0.648 (Zhangjiakou) to 0.922 (Baoding) and O_3 predictions from 0.834 (Qinhuangdao) to 0.901 (Tianjin). Figures S3 and S4 display scatter plots and regression results of predicted versus observed values for the cities with the highest and lowest forecasting accuracies for $PM_{2.5}$ and O_3 using Models I and II, respectively. Although prediction accuracies varied across cities, both models proved effective in forecasting pollutant concentrations, providing accurate estimations for most observations. The seasonal performance of Models I and II for $PM_{2.5}$ and O_3 predictions is illustrated in Figures S5–S8. The models demonstrated relatively stable performance for O_3 predictions across seasons, while $PM_{2.5}$ predictions showed some variability. Notably, the models tended to underestimate $PM_{2.5}$ concentrations during severely polluted events in spring, possibly due to unique emission sources such as agricultural burning (e.g., crop straw and biomass), which are challenging to quantify due to their fragmented and unregulated nature. Incomplete or outdated emission inventories may contribute to these underestimations. To assess model robustness, the independent variables in the original dataset were doubled and increased by 10%. The results showed no significant changes in R^2 , RMSE, or MAE, indicating strong model robustness and generalizability.

Table 4. Error analysis of $PM_{2.5}$ and O_3 predictions using XGBoost across cities in the Beijing–Tianjin–Hebei region.

City	Pollutant	Model I						Model II					
		RMSE		MAE		R2		RMSE		MAE		R2	
		Training	Testing	Training	Testing	Training	Testing	Training	Testing	Training	Testing	Training	Testing
Beijing	$PM_{2.5}$	3.04	8.69	2.15	4.97	0.993	0.945	7.03	15.66	4.79	10.25	0.962	0.807
	O_3	4.64	11	3.24	7.62	0.99	0.947	7.04	15.17	5.02	10.82	0.978	0.896
Tianjin	$PM_{2.5}$	3.39	8.54	2.36	5.37	0.993	0.953	8.09	17.42	5.59	11.88	0.958	0.799
	O_3	5.26	12.9	3.61	8.75	0.989	0.933	7.43	15.82	5.23	11.2	0.978	0.901
Shijiazhuang	$PM_{2.5}$	2.67	6.98	1.89	4.24	0.996	0.976	7.39	16.85	5.08	11.05	0.972	0.851
	O_3	5.36	12.9	3.7	8.72	0.989	0.936	7.16	16.55	5.09	11.81	0.980	0.895
Baoding	$PM_{2.5}$	3.28	9.41	2.38	5.22	0.995	0.958	5.23	12.42	3.67	8.05	0.987	0.922
	O_3	5.69	13.5	3.96	9.33	0.989	0.935	8.5	18.68	6.04	13.58	0.974	0.877
Tangshan	$PM_{2.5}$	3.25	8.42	2.34	5.31	0.993	0.954	8.63	19.51	5.93	12.88	0.952	0.741
	O_3	6.18	15.3	4.16	10.3	0.985	0.911	8.71	19.17	6.17	13.53	0.971	0.856
Qinhuangdao	$PM_{2.5}$	2.68	6.43	1.94	4.08	0.993	0.958	7.04	15.56	4.72	10.14	0.950	0.768
	O_3	5.74	12.9	4.06	9.19	0.984	0.918	8.68	18.50	6.25	13.50	0.964	0.834
Handan	$PM_{2.5}$	3.92	9.63	2.84	6.34	0.993	0.954	9.57	22.02	6.65	14.49	0.959	0.763
	O_3	6.02	14.5	4.25	10.2	0.987	0.923	8.14	18.15	5.82	13.10	0.976	0.875
Zhangjiakou	$PM_{2.5}$	2.85	7.61	2.09	3.89	0.989	0.946	4.62	18.31	3.14	7.17	0.971	0.648
	O_3	5.79	12.3	4.18	9	0.974	0.886	6.37	13.70	4.49	9.78	0.969	0.856
Chengde	$PM_{2.5}$	2.78	7.44	2.04	4.07	0.987	0.918	5.57	11.55	3.77	7.48	0.948	0.751
	O_3	5.63	13.3	3.88	9.19	0.985	0.916	8.28	17.33	5.84	12.32	0.966	0.856
Langfang	$PM_{2.5}$	3.14	12.1	2.32	5.01	0.994	0.914	8.67	18.52	5.95	12.16	0.952	0.772
	O_3	5.99	14.1	4.11	9.59	0.987	0.927	9.48	19.64	6.79	14.14	0.967	0.859
Cangzhou	$PM_{2.5}$	3.29	7.99	2.39	5.09	0.993	0.96	8.36	18.77	5.74	12.30	0.957	0.777
	O_3	6.21	14.3	4.33	10.2	0.984	0.915	8.21	17.77	5.89	13.28	0.973	0.872
Hengshui	$PM_{2.5}$	3.65	8.45	2.68	5.76	0.993	0.963	9.03	19.97	6.24	13.21	0.958	0.785
	O_3	6.04	13.8	4.23	9.89	0.986	0.929	8.13	17.46	5.82	12.87	0.974	0.879
Xingtai	$PM_{2.5}$	4.08	9.31	2.94	6.25	0.993	0.959	8.75	21.31	6.08	14.07	0.965	0.790
	O_3	6.41	14.9	4.49	10.4	0.986	0.925	8.75	19.21	6.29	14.10	0.974	0.873

Table 5 compares the model fitting and validation results of this study with other relevant studies conducted in China, offering valuable insights into the performance of the models used. For $PM_{2.5}$ estimation, our XGBoost-based models outperformed STLG [59], Space-Time Extra Trees [60], LME [61], and RF [62] reported in prior studies. For O_3 estimation, our models surpassed RF [63] and achieved accuracy comparable to WRFC-XGB [64].

Table 5. Comparison of model performance and validation outcomes with recent studies conducted in various regions.

Study Area	Pollutant	Spatiotemporal Resolution	Model	Model Fitting (Training Dataset)		Cross-Validation (Testing Dataset)			Source
				R ²	RMSE (µg/m ³)	Method	R ²	RMSE (µg/m ³)	
China	PM _{2.5}	1 km (monthly)	RF	-	-	10-fold	0.88	14.89	(Huang et al., 2018) [59]
China	PM _{2.5}	1 km (daily)	Space-time extra trees	0.92–0.94	5.11–9.92	10-fold	0.86–0.90	10.00–18.40	(Wei et al., 2021) [60]
China	PM _{2.5}	5 km (hourly)	STLG	0.97–0.98	4.18–7.31	10-fold	0.81–0.85	11.24–15.56	(Wei et al., 2021) [61]
Taiwan	PM _{2.5}	10 km (twice-daily)	LME	0.77	11.4	10-fold	0.66	12.9	(Jung et al., 2018) [62]
BTH	PM _{2.5}	5 km (hourly)	XGBoost	0.98–0.99	2.67–4.08	10-fold	0.91–0.98	6.43–12.1	Our study (Model I)
BTH	PM _{2.5}	5 km (hourly)	XGBoost	0.95–0.99	4.62–9.57	10-fold	0.65–0.92	11.55–22.02	Our study (Model II)
BTH	O ₃	Daily (MDA8H)	RF	-	-	10-fold	0.84		(Ma et al., 2021) [63]
BTH	O ₃	Daily	WRFC-XGB	0.91–0.95	15.50–17.7	10-fold	0.91–0.95	13.57–17.7	(Hu et al., 2022) [64]
BTH	O ₃	5 km (hourly)	XGBoost	0.97–0.99	4.64–6.41	10-fold	0.89–0.95	11–15.3	Our study (Model I)
BTH	O ₃	5 km (hourly)	XGBoost	0.96–0.98	6.37–9.48	10-fold	0.83–0.90	13.7–19.64	Our study (Model II)

BTH indicates Beijing–Tianjin–Hebei region.

3.3. Major Impact Factors of PM_{2.5} and O₃ Predictions

Figure 5a illustrates the variable importance scores for PM_{2.5} concentrations derived from Model I. Across all investigated cities, PM₁₀ and CO exhibited the most significant impacts on PM_{2.5}, with importance scores exceeding 10%. This is likely due to their shared emission sources and similar dispersion characteristics in the atmosphere. Meteorological parameters showed lower importance scores, possibly because their influence is overshadowed by PM₁₀ and CO. Figure 5b presents the results from Model II, which differ markedly from those of Model I. The lagged PM_{2.5} concentration ($t - 1$) consistently had the highest importance score across all cities. Meteorological factors, such as temperature and wind speed, also demonstrated significant impacts, with relatively stable importance scores across different cities and seasons. Notably, the Baoding region stands out, with CO exhibiting a particularly high importance score.

Figure 5c presents the variable importance scores for O₃ concentration predictions using Model I. The results consistently indicate that temperature and NO₂ concentration had the highest importance scores among all parameters, underscoring their critical roles in O₃ formation. Under sunlight, nitrogen oxides (NO_x) and volatile organic compounds (VOCs) undergo photochemical reactions to produce secondary pollutants, including O₃. Higher temperatures enhance O₃ formation by accelerating these reactions. Notably, the relative importance of temperature and NO₂ varied by season: temperature was the dominant factor in spring, summer, and autumn, while the NO₂ concentration became the most significant parameter in winter. Additionally, PM₁₀ concentration showed high importance scores in summer and autumn, likely because increased PM₁₀ levels reduce visibility, suppressing O₃ generation. Figure 5d illustrates the variable importance scores for O₃ concentration forecasts using Model II. Apart from the lagged O₃ concentration ($t - 1$), temperature consistently exhibited the highest scores across most cities and seasons, highlighting its pivotal role in O₃ formation. Temperature and surface radiation are key drivers of O₃ production due to their influence on atmospheric photochemical reactions. Higher temperatures accelerate the reaction rates of O₃ precursors, such as VOCs and NO_x, while also promoting VOC evaporation, thereby increasing their atmospheric concentrations. Surface radiation, particularly ultraviolet (UV) radiation from sunlight, provides the energy required for NO₂ photolysis, producing reactive oxygen radicals that combine with molecular oxygen to form O₃. The intensity of this radiation directly affects the rate of O₃

production. These mechanisms are well-documented in the atmospheric chemistry and physics literature [59].

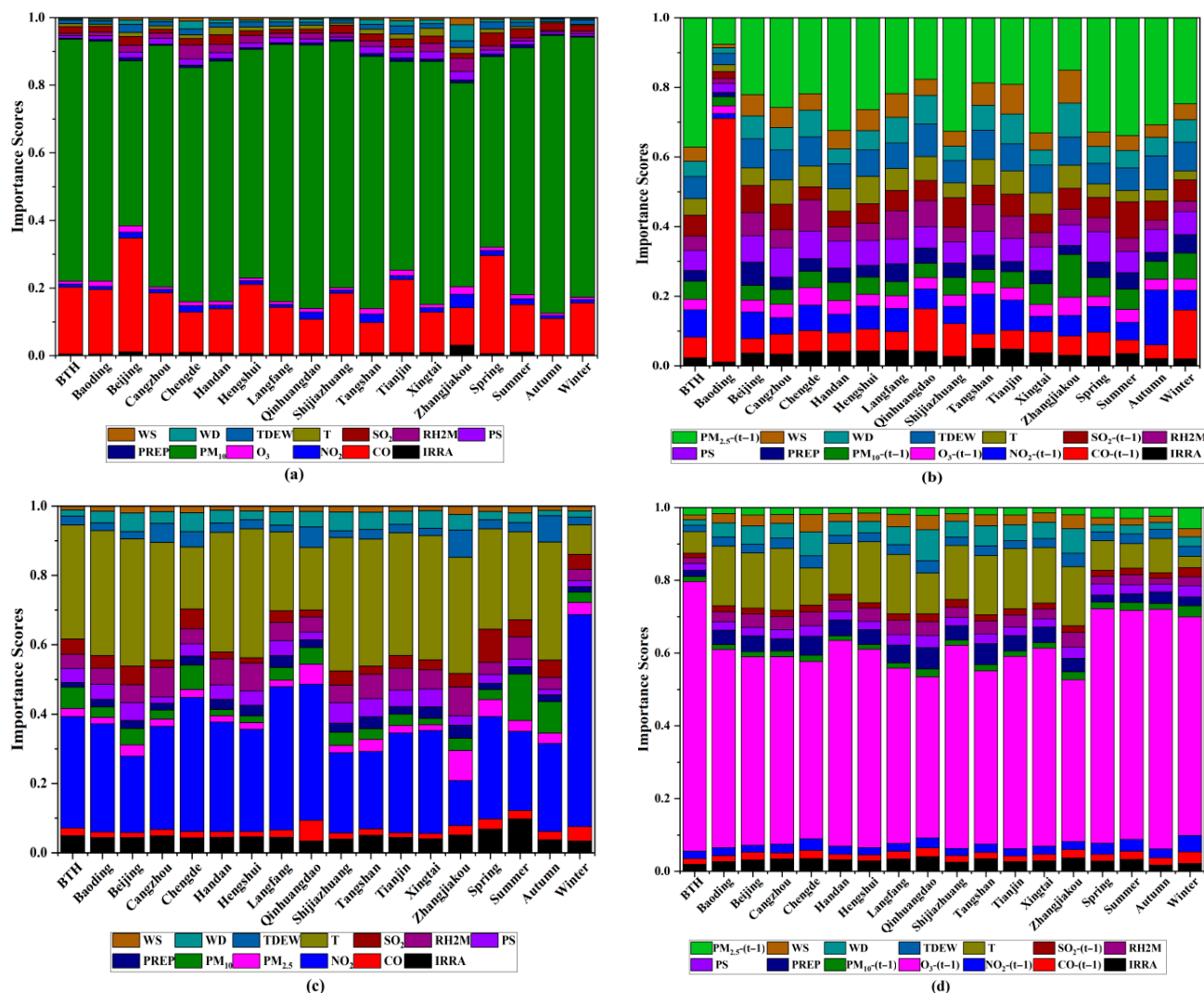


Figure 5. The importance scores of variables on pollutant prediction for different cities and seasons: (a) Model I for $PM_{2.5}$ concentration; (b) Model II for $PM_{2.5}$ concentration; (c) Model I for O_3 concentration; (d) Model II for O_3 concentration.

Other meteorological factors, including precipitation, wind, and solar radiation, along with pollutant concentrations such as $PM_{2.5}$ and NO_2 , significantly influenced O_3 concentrations. The ranking of variable importance scores varied across cities and seasons. Notably, in winter, $PM_{2.5}$ and NO_2 concentrations exhibited higher importance scores than temperature.

Generally, Models I and II showed notable differences in predicting $PM_{2.5}$ and O_3 . Model I identified strong associations between $PM_{2.5}$ and PM_{10} , as well as CO, which can be attributed to the impacts of anthropogenic emissions. In contrast, Model II suggested that both meteorological and anthropogenic emission factors influenced $PM_{2.5}$ concentrations, with the main contributing factors varying across different cities. The primary reason for these differences may lie in the pollutant lag incorporated in Model II. Meteorological factors tend to affect the diffusion and transformation of pollutants in similar ways, and as a result, their influence on $PM_{2.5}$ could be overshadowed by the spatial and temporal distribution patterns of other pollutants, such as PM_{10} and CO. For this reason, Model

I did not capture seasonal changes in the main influencing factors for PM_{2.5} prediction, while Model II did. Similarly, the two models differed in predicting O₃ concentrations. Model I identified NO₂ and temperature as the primary influencing factors, while Model II found that the main factors varied by city and season. When predicting PM_{2.5} and O₃ using lagged anthropogenic emission factors, it is crucial to consider both anthropogenic and meteorological influences across different cities and seasons.

From the SHAP values of different influencing factors in Figure S9, Model I demonstrates that PM₁₀ has the greatest impact on PM_{2.5} prediction and NO₂ has the greatest impact on O₃ prediction. For Model II, SO₂ has the greatest impact on PM_{2.5} prediction and T has the greatest impact on O₃ prediction.

3.4. Policy Implications

Suggestions for Different Seasons

1. Strengthening Winter Pollution Prevention and Control

Winter is the season with the highest PM_{2.5} concentrations, which remained elevated from 2019 to 2021. It is recommended to strictly control the combustion of loose coal, promote clean heating alternatives, and increase subsidies for “coal-to-electricity” and “coal-to-gas” transitions in rural areas. Additionally, emission reductions from industrial sources should be intensified, particularly for key industries such as steel and building materials. Stricter production restrictions and emission reduction measures should be implemented during winter, supported by real-time supervision through online monitoring systems. Traffic control should also be optimized by encouraging the use of public transportation and new energy vehicles during peak hours, and imposing regional traffic restrictions on high-emission vehicles to reduce the contribution of vehicle exhausts.

2. Consolidating Summer Pollution Prevention and Control

Although summer typically has lower PM_{2.5} concentrations, the concentration in summer 2023 reached 20.69 µg/m³, highlighting the need to consolidate pollution control efforts. Coordinated control of volatile organic compounds (VOCs) and nitrogen oxides should be strengthened, as high summer temperatures promote O₃ formation, and VOCs are also key precursors for PM_{2.5}. Efforts should focus on controlling VOC emissions from industries like coatings and printing, while simultaneously addressing nitrogen oxide emissions from industrial furnaces and power plants. Meteorological conditions, such as rain and favorable diffusion, should be leveraged to optimize control measures. Emergency emission reduction plans should be developed in advance to respond quickly at the early stages of pollution events, keeping concentrations at lower levels.

3. Responding to Interannual Fluctuations

The PM_{2.5} concentration in 2023 showed a rebound compared with 2022 (from 37.4 to 40.5 µg/m³). It is crucial to dynamically adjust emission reduction strategies and establish a “pollution rebound early warning–source tracing–response” mechanism. In the event of a sustained increase in concentration, rapid investigations should be conducted to identify changes in pollution sources, such as the resumption of industrial production or the re-emission of loose coal. Specific controls should be tightened accordingly. Regional joint prevention and control efforts must be intensified, with the Beijing–Tianjin–Hebei region and surrounding areas acting in unison. Unifying standards for winter production restrictions and emergency emission reduction measures will help avoid concentration rebounds caused by cross-regional pollution transmission.

3.5. Suggestions Based on PM_{2.5} and O₃ Concentration Differences Among Cities

1. For PM_{2.5}-Focused Cities (Handan, Xingtai, and Shijiazhuang)

Heavy-industry regulation: These cities should strengthen the transformation of high-pollution industries, such as steel and cement production. Stricter emission standards should be set, advanced pollution-reduction technologies promoted, and phased production cuts scheduled during high-pollution risk periods.

Coal-related emissions reduction: Given the significant contribution of coal combustion to PM_{2.5}, accelerate the shift to clean energy in both industrial and domestic sectors. Strengthen oversight of small-scale coal-fired facilities to ensure the elimination of substandard emissions.

Green barrier construction: Increase the construction of urban green belts, particularly in industrial zones and around transportation hubs. Use vegetation to capture particulate matter and enhance air quality.

Dust management: Strengthen road cleaning and dust suppression measures. Impose stricter dust-control requirements on construction sites, mandating the use of dust-suppression techniques such as enclosures and sprinklers.

2. For Low-PM_{2.5} Cities (Chengde and Zhangjiakou)

Ecological conservation: Maintain and further enhance existing ecological advantages. Expand forest coverage and wetland areas to bolster the natural air purification capacity.

Model promotion: Share successful experiences in ecological protection and low-pollution development with other cities. Participate in regional environmental cooperation projects to serve as demonstration areas for air-quality management.

3. For High-O₃ Cities (Cangzhou, Hengshui, and Xingtai)

Industry-specific governance: Focus on industries with high VOC emissions, such as coating and printing. Promote the use of low-VOC raw materials and advanced treatment technologies. Strengthen control over NO_x emissions from industrial boilers and vehicles.

Monitoring and early warning: Establish a sophisticated O₃ monitoring and early-warning system. Predict high-O₃ risk periods based on meteorological conditions and implement temporary emission-reduction measures in advance, such as adjusting industrial production schedules and imposing vehicle travel restrictions.

Data-sharing platform: Create a regional air-quality data-sharing platform to ensure the timely exchange of monitoring data between cities. Facilitate joint analysis of pollution sources and transmission paths.

Coordinated action plan: Develop a unified regional air-pollution control action plan. During extreme pollution events, cities should collaborate on implementing emergency measures, such as unified vehicle restrictions and industrial production adjustments, to prevent cross-city pollution transmission.

Future research should focus on improving model interpretability to gain a better understanding of the complex interactions between variables and pollutant formation mechanisms. Incorporating additional data sources, such as real-time traffic flow, industrial activity, and satellite remote sensing, could enhance both model accuracy and spatial resolution. Furthermore, integrating chemical transport models with machine learning approaches may offer deeper insights into pollutant dynamics and atmospheric chemistry. Long-term monitoring and validation across diverse geographic regions will be crucial for generalizing the applicability of the models. Addressing these challenges will strengthen air pollution prediction capabilities and support more effective environmental policies and management strategies.

4. Conclusions

This study investigates the spatiotemporal distribution of PM_{2.5} and O₃ in the Beijing–Tianjin–Hebei (BTH) region and applies machine learning algorithms for concentration prediction, leading to the following key findings.

1. Spatiotemporal Patterns of Pollutants

Seasonal variations: PM_{2.5} shows a decreasing trend from 2019 to 2023 (49.39 to 37.4 µg/m³), with a rebound in 2023 (40.5 µg/m³). It peaks in winter (up to 79.79 µg/m³) due to heating emissions and stable meteorological conditions and reaches its lowest levels in summer (20.69 µg/m³). O₃ exhibits stable annual mean concentrations (63.45–69.84 µg/m³), with the highest levels in summer (due to photochemical reactions) and the lowest in winter (due to cold-induced inhibition). Spatial differences: PM_{2.5} concentrations are highest in Handan, Xingtai, and Shijiazhuang, while Chengde and Zhangjiakou show lower levels. Spatial variations in O₃ are less pronounced, but higher concentrations are observed in Shijiazhuang, Hengshui, and Xingtai.

2. Machine Learning Models for Pollution Prediction

Model performance: XGBoost outperforms both FCNN and RF in predicting PM_{2.5} and O₃, with testing R² values of 0.979 (for PM_{2.5}) and 0.938 (for O₃), demonstrating its strong capability in capturing spatiotemporal pollution patterns. Model I (without lagged variables) is highly effective in associating pollutants with emission sources (e.g., PM₁₀ and CO for PM_{2.5}), while Model II (with lagged variables) emphasizes meteorological impacts (temperature and wind speed) on pollutant diffusion. Applicability: Both models effectively forecast pollutant concentrations across cities, though Model II shows lower accuracy due to the complex interactions between emissions and meteorological factors.

3. Key Influencing Factors

PM_{2.5}: PM₁₀ and CO are the dominant drivers (with importance >10%) due to shared emission sources (e.g., fuel combustion and industrial processes). Meteorological factors such as temperature and wind speed play a significant role in Model II, with variations across cities (e.g., CO in Baoding). O₃: Temperature and NO₂ are key contributors, with temperature being more influential during warm seasons and NO₂ during winter. PM₁₀ also suppresses O₃ formation during summer and autumn by reducing visibility.

4. Policy Implications and Future Directions

Seasonal and urban-specific recommendations are as follows. Winter: Strengthen clean heating solutions, enforce industrial emission reductions, and improve traffic control measures. Summer: Co-manage VOCs and NO_x emissions to mitigate O₃ formation. High-PM_{2.5} cities: Focus on regulating heavy industries and reducing coal combustion. Low-PM_{2.5} cities: Promote ecological conservation and natural air purification. Future research: Focus on improving model interpretability, integrating real-time data (e.g., traffic and industrial activity), and combining machine learning techniques with chemical transport models to enhance prediction accuracy and support policymaking.

In summary, this study provides empirical evidence on pollution dynamics and introduces robust prediction tools that can support targeted air quality management strategies in the BTH region.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/su17209211/s1>, Table S1. Parameter setting of the models using different machine learning algorithms. Figure S1: The performance of XGBoost (a), RF (b), and FCNN (c) models in training and predicting PM_{2.5} and O₃ (Model I). Figure S2: The performance of XGBoost (a), RF (b), and FCNN (c) models in training and predicting PM_{2.5} and O₃ (Model II).

Figure S3. XGBoost prediction of PM_{2.5} concentrations in Shijiazhuang and Langfang areas, and O₃ concentrations in Beijing and Zhangjiakou areas (Model I, testing dataset). Figure S4. XGBoost prediction of PM_{2.5} concentrations in Beijing and Zhangjiakou areas, and O₃ concentrations in Beijing and Qinhuangdao areas (Model II, testing dataset). Figure S5. XGBoost's performance in predicting PM_{2.5} levels in spring, summer, autumn, and winter (Model I). Figure S6. XGBoost's performance in predicting O₃ levels in spring, summer, autumn, and winter (Model I). Figure S7. XGBoost's performance in predicting PM_{2.5} levels in spring, summer, autumn, and winter (Model II). Figure S8. XGBoost's performance in predicting O₃ levels in spring, summer, autumn, and winter (Model II). Figure S9. The SHAP value of variables on pollutant prediction for BTH: (a) Model I for PM_{2.5} concentration; and (b) Model I for O₃ concentration; (c) Model II for PM_{2.5} concentration; (d) Model II for O₃ concentration. References [65–67] are cited in Supplementary Materials.

Author Contributions: Conceptualization: Y.H.; Methodology: C.W.; Software: C.W.; Validation: C.Z.; Formal Analysis: C.Z.; Investigation: C.W.; Resources: C.W.; Data Curation: C.W.; Writing—Original Draft Preparation: C.W.; Writing—Review and Editing: Y.H.; Visualization: Y.T.; Supervision: Y.T.; Project Administration: C.Z.; Funding Acquisition: C.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the National Key R&D Program of China (2022YFD1601001).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data will be made available on request.

Acknowledgments: We would like to thank NASA for providing the meteorological data through the POWER (Prediction of Worldwide Energy Resource) project (<https://power.larc.nasa.gov/data-access-viewer>, accessed on 1 February 2024), which was essential for our analysis. We are also grateful to the reviewers for their constructive comments and suggestions that greatly improved this manuscript.

Conflicts of Interest: The authors declare no conflicts of interest.

References

- Li, C.; van Donkelaar, A.; Hammer, M.S.; McDuffie, E.E.; Burnett, R.T.; Spadaro, J.V.; Chatterjee, D.; Cohen, A.J.; Apte, J.S.; Southerland, V.A.; et al. Reversal of trends in global fine particulate matter air pollution. *Nat. Commun.* **2023**, *14*, 5349. [CrossRef]
- Zhao, X.; Zhang, Z.; Xu, J.; Gao, J.; Cheng, S.; Zhao, X.; Xia, X.; Hu, B. Impacts of aerosol direct effects on PM_{2.5} and O₃ respond to the reductions of different primary emissions in Beijing-Tianjin-Hebei and surrounding area. *Atmos. Environ.* **2023**, *309*, 119948. [CrossRef]
- Wang, S.; Ren, Y.; Xia, B. PM_{2.5} and O₃ concentration estimation based on interpretable machine learning. *Atmos. Pollut. Res.* **2023**, *14*, 101866. [CrossRef]
- Zeng, Q.; Li, Y.; Tao, J.; Fan, M.; Chen, L.; Wang, L.; Wang, Y. Full-coverage estimation of PM_{2.5} in the Beijing-Tianjin-Hebei region by using a two-stage model. *Atmos. Environ.* **2023**, *309*, 119956. [CrossRef]
- Luo, Z.; Lu, P.; Chen, Z.; Liu, R. Ozone Concentration Estimation and Meteorological Impact Quantification in the Beijing-Tianjin-Hebei Region Based on Machine Learning Models. *Earth Space Sci.* **2024**, *11*, e2023EA003346. [CrossRef]
- Cao, J.J.; Lee, S.C.; Ho, K.F.; Fung, K. Characteristics, sources, and health impacts of atmospheric particulate matter in China. *Sci. Total Environ.* **2019**, *659*, 400–413.
- Wang, G.H.; Zhang, Y.H.; Zhang, Q.; Zheng, B.; He, K.B.; Cofala, J. China's anthropogenic sulfur dioxide, nitrogen oxides, and primary fine particulate matter emissions, 1990–2015: A high-resolution emission inventory. *Atmos. Chem. Phys.* **2017**, *17*, 11925–11952.
- Liu, F.; Zhang, Y.; Zhang, Q.; Wang, S.; He, K. Vehicle emission control in China: Review and outlook. *Sci. Total Environ.* **2018**, *619–620*, 1195–1208. [CrossRef]
- Li, X.; Zhang, Q.; He, K.B.; Zheng, B.; Streets, D.G. Anthropogenic mercury emissions in China during 1990–2010: A provincial-level inventory. *Atmos. Chem. Phys.* **2016**, *16*, 15609–15622.
- Crutzen, P.J. The role of NO and NO₂ in the chemistry of the troposphere and stratosphere. *Ann. N. Y. Acad. Sci.* **1979**, *322*, 16–47. [CrossRef]

11. Lin, Y.; Zhang, Q.; Streets, D.G.; He, K.B.; Wang, S.X. Ozone trends in China from 1991 to 2012: Analysis of surface observations and model simulations. *Atmos. Chem. Phys.* **2014**, *14*, 5819–5834.
12. Pope, C.A.; Dockery, D.W. Health effects of fine particulate air pollution: Lines that connect. *J. Air Waste Manag. Assoc.* **2006**, *56*, 709–742. [[CrossRef](#)]
13. Brook, R.D.; Rajagopalan, S.; Pope, C.A., 3rd; Brook, J.R.; Bhatnagar, A.; Diez-Roux, A.V.; Holguin, F.; Hong, Y.; Luepker, R.V.; Mittleman, M.A.; et al. Particulate matter air pollution and cardiovascular disease: An update to the scientific statement from the American Heart Association. *Circulation* **2010**, *121*, 2331–2378. [[CrossRef](#)]
14. Krewski, D.; Jerrett, M.; Burnett, R.T.; Ma, R.; Hughes, E.; Shi, Y.; Turner, M.C.; Pope, C.A., 3rd; Thurston, G.; Calle, E.E.; et al. *Extended Follow-Up and Spatial Analysis of the American Cancer Society Study Linking Particulate Air Pollution and Mortality*; Research Report; Health Effects Institute: Boston, MA, USA, 2009; pp. 5–86, discussion 115–136.
15. Wang, S.; Zhang, Q.; Zheng, B.; Cofala, J.; He, K.B. Emission trends and future projections of air pollutants in China: Implications for air quality improvement. *Atmos. Chem. Phys.* **2023**, *23*, 3229–3246.
16. Atkinson, R.; Arey, J. Atmospheric degradation of volatile organic compounds. *Chem. Rev.* **2003**, *103*, 4605–4638. [[CrossRef](#)]
17. Bell, M.L.; McDermott, A.; Zeger, S.L.; Samet, J.M.; Dominici, F. Ozone and short-term mortality in 95 US urban communities, 1987–2000. *JAMA* **2006**, *295*, 1087–1095. [[CrossRef](#)]
18. Zheng, J.; Zhang, Q.; He, K.B.; Wang, S.; Streets, D.G. Tropospheric ozone in China: Concentrations, trends, and sources. *Atmos. Chem. Phys.* **2021**, *21*, 13239–13258.
19. Huang, Z.; Zhong, Z.; Sha, Q.; Xu, Y.; Zhang, Z.; Wu, L.; Wang, Y.; Zhang, L.; Cui, X.; Tang, M.; et al. An updated model-ready emission inventory for Guangdong Province by incorporating big data and mapping onto multiple chemical mechanisms. *Sci. Total Environ.* **2021**, *769*, 144535. [[CrossRef](#)] [[PubMed](#)]
20. Ren, X.; Mi, Z.; Georgopoulos, P.G. Comparison of Machine Learning and Land Use Regression for fine scale spatiotemporal estimation of ambient air pollution: Modeling ozone concentrations across the contiguous United States. *Environ. Int.* **2020**, *142*, 105827. [[CrossRef](#)]
21. Abdullah, S.; Ismail, M.; Ahmed, A.N. Identification of air pollution potential sources through principal component analysis (PCA). *Int. J. Civ. Eng. Technol.* **2018**, *9*, 1435–1442.
22. Jain, A.; Lella, R.L. Pearson correlation coefficient based attribute weighted k-nn for air pollution prediction. In Proceedings of the 2020 IEEE 17th India Council International Conference (INDICON), New Delhi, India, 10–13 December 2020; pp. 1–8.
23. Hoek, G.; Beelen, R.; de Hoogh, K.; Vienneau, D.; Gulliver, J.; Fischer, P.; Briggs, D. A review of land-use regression models to assess spatial variation of outdoor air pollution. *Atmos. Environ.* **2008**, *42*, 7561–7578. [[CrossRef](#)]
24. Zhou, Q.; Wang, C.; Fang, S. Application of geographically weighted regression (GWR) in the analysis of the cause of haze pollution in China. *Atmos. Pollut. Res.* **2019**, *10*, 835–846. [[CrossRef](#)]
25. Han, Y.; Zhang, Q.; Li, V.O.K.; Lam, J.C.K. Deep-AIR: A Hybrid CNN-LSTM Framework for Air Quality Modeling in Metropolitan Cities. *arXiv* **2021**, arXiv:2103.14587.
26. Hu, X.; Belle, J.H.; Meng, X.; Wildani, A.; Waller, L.A.; Strickland, M.J.; Liu, Y. Estimating PM_{2.5} Concentrations in the Conterminous United States Using the Random Forest Approach. *Environ. Sci. Technol.* **2017**, *51*, 6936–6944. [[CrossRef](#)]
27. Zhang, K.; Batterman, Z.S. Air pollution and health risks due to vehicle traffic. *Sci. Total Environ.* **2013**, *450–451*, 307–316. [[CrossRef](#)]
28. Zhang, K.; Batterman, S. Air pollution prediction using XGBoost with multiple spatiotemporal features. *Atmos. Pollut. Res.* **2019**, *10*, 768–775. [[CrossRef](#)]
29. Jogin, M.; Mohana; Madhulika, M.S.; Divya, G.D.; Meghana, R.K.; Apoorva, S. Feature Extraction using Convolution Neural Networks (CNN) and Deep Learning. In Proceedings of the 2018 3rd IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT), Bangalore, India, 18–19 May 2018; pp. 2319–2323.
30. Zhou, C.; Wang, F.; Guo, Y.; Liu, C.; Ji, D.; Wang, Y.; Xu, X.; Lu, X.; Wang, Y.; Carmichael, G.; et al. Reconstructed daily ground-level O₃ in China over 2005–2021 for climatological, ecological, and health research. *Earth Syst. Sci. Data Discuss.* **2022**. [[CrossRef](#)]
31. Gui, K.; Che, H.; Zeng, Z.; Wang, Y.; Zhai, S.; Wang, Z.; Luo, M.; Zhang, L.; Liao, T.; Zhao, H.; et al. Construction of a virtual PM_{2.5} observation network in China based on high-density surface meteorological observations using the Extreme Gradient Boosting model. *Environ. Int.* **2020**, *141*, 105801. [[CrossRef](#)]
32. Yin, J.; Jiang, J.; Tong, L.; Huang, P. FCNN+: An Improved Fully Connected Neural Network High-accuracy Prediction Model. In Proceedings of the 2023 8th International Conference on Information Systems Engineering (ICISE), Dalian, China, 23–25 June 2023; pp. 539–542. [[CrossRef](#)]
33. Tian, H.; Zhao, Y.; Luo, M.; He, Q.; Han, Y.; Zeng, Z. Estimating PM_{2.5} from multisource data: A comparison of different machine learning models in the Pearl River Delta of China. *Urban Clim.* **2021**, *35*, 100740. [[CrossRef](#)]
34. Murphy, K.P. *Machine Learning: A Probabilistic Perspective*; MIT Press: Cambridge, MA, USA, 2012.
35. Gupta, P.; Zhan, S.; Mishra, V.; Aekakkararungroj, A.; Markert, A.; Paibong, S.; Chishtie, F. Machine learning algorithm for estimating surface PM_{2.5} in Thailand. *Aerosol Air Qual. Res.* **2021**, *21*, 210105. [[CrossRef](#)]

36. Vu, T.V.; Shi, Z.; Cheng, J.; Zhang, Q.; He, K.; Wang, S.; Harrison, R.M. Assessing the impact of clean air action on air quality trends in Beijing using a machine learning technique. *Atmos. Chem. Phys.* **2019**, *19*, 11303–11314. [\[CrossRef\]](#)
37. Zhao, J.; Deng, F.; Cai, Y.; Chen, J. Long short-term memory-Fully connected (LSTM-FC) neural network for PM_{2.5} concentration prediction. *Chemosphere* **2019**, *220*, 486–492. [\[CrossRef\]](#) [\[PubMed\]](#)
38. Rauschmayr, N.; Kumar, V.; Huilgol, R.; Olgiati, A.; Bhattacharjee, S.; Harish, N.; Kenthapadi, K. Amazon sagemaker debugger: A system for real-time insights into machine learning model training. *Proc. Mach. Learn. Syst.* **2021**, *3*, 770–782.
39. Takoutsing, B.; Heuvelink, G.B.M. Comparing the prediction performance, uncertainty quantification and extrapolation potential of regression kriging and random forest while accounting for soil measurement errors. *Geoderma* **2022**, *428*, 116192. [\[CrossRef\]](#)
40. Vaysse, K.; Lagacherie, P. Using quantile regression forest to estimate uncertainty of digital soil mapping products. *Geoderma* **2017**, *291*, 55–64. [\[CrossRef\]](#)
41. Xiao, Z.; Li, H.; Gao, Y. Analysis of the impact of the Beijing-Tianjin-Hebei coordinated development on environmental pollution and its mechanism. *Environ. Monit. Assess* **2022**, *194*, 91. [\[CrossRef\]](#) [\[PubMed\]](#)
42. Monks, P.S.; Archibald, A.T.; Colette, A.; Cooper, O.; Coyle, M.; Derwent, R.; Fowler, D.; Granier, C.; Law, K.S.; Mills, G.E.; et al. Tropospheric ozone and its precursors from the urban to the global scale from air quality to short-lived climate forcer. *Atmos. Chem. Phys.* **2015**, *15*, 8889–8973. [\[CrossRef\]](#)
43. Jacob, D.J.; Winner, D.A. Effect of climate change on air quality. *Atmos. Environ.* **2009**, *43*, 51–63. [\[CrossRef\]](#)
44. Zhang, Q.; Jimenez, J.L.; Canagaratna, M.R.; Ulbrich, I.M.; Ng, N.L.; Worsnop, D.R.; Sun, Y. Understanding atmospheric organic aerosols via factor analysis of aerosol mass spectrometry: A review. *Anal. Bioanal. Chem.* **2011**, *401*, 3045–3067. [\[CrossRef\]](#)
45. Arya, S.P. *Air Pollution Meteorology and Dispersion*; Oxford University Press: Oxford, UK, 1999.
46. Han, L.; Lan, T.; Cheng, S.; Wang, Y.; Qi, C.; Tian, J.; Wang, H.; Han, D.; Wang, S. Evolution Characteristics of PM_{2.5} and O₃ and Their Synergistic Effects on Atmospheric Compound Pollution in Tangshan. *Environ. Sci.* **2024**, *45*, 4385–4397.
47. Alves, C.; Evtyugina, M.; Vicente, E.; Vicente, A.; Rienda, I.C.; de la Campa, A.S.; Duarte, I. PM_{2.5} chemical composition and health risks by inhalation near a chemical complex. *J. Environ. Sci.* **2023**, *124*, 860–874. [\[CrossRef\]](#)
48. Zeng, J.; Zhang, L.; Yao, C.; Xie, T.; Rao, L.; Lu, H.; Lu, S. Relationships between chemical elements of PM_{2.5} and O₃ in Shanghai atmosphere based on the 1-year monitoring observation. *J. Environ. Sci.* **2020**, *95*, 49–57. [\[CrossRef\]](#) [\[PubMed\]](#)
49. Seinfeld, J.H.; Pandis, S.N. *Atmospheric Chemistry and Physics: From Air Pollution to Climate Change*; John Wiley & Sons: Hoboken, NJ, USA, 2016.
50. Ervens, B.; Turpin, B.J.; Weber, R.J. Secondary organic aerosol formation in cloud droplets and aqueous particles (aqSOA): A review of laboratory, field and model studies. *Atmos. Chem. Phys.* **2011**, *11*, 11069–11129. [\[CrossRef\]](#)
51. Tao, J.; Zhang, L.; Cao, J.; Zhong, L.; Chen, D.; Yang, Y.; Zhang, R. Source apportionment of PM_{2.5} at urban and suburban areas of the Pearl River Delta region, south China-With emphasis on ship emissions. *Sci. Total Environ.* **2017**, *574*, 1559–1570. [\[CrossRef\]](#)
52. Liu, L.; Long, X.; Li, Y.; Zang, Z.; Wang, F.; Han, Y.; Yang, J. Impacts of meteorology and emission reductions on haze pollution during the lockdown in the North China Plain. *Atmos. Chem. Phys.* **2025**, *25*, 1569–1585. [\[CrossRef\]](#)
53. Shi, C.; Yuan, R.; Wu, B.; Meng, Y.; Zhang, H.; Zhang, H.; Gong, Z. Meteorological conditions conducive to PM_{2.5} pollution in winter 2016/2017 in the Western Yangtze River Delta, China. *Sci. Total Environ.* **2018**, *642*, 1221–1232. [\[CrossRef\]](#)
54. Bhatti, U.A.; Yan, Y.; Zhou, M.; Ali, S.; Hussain, A.; Qingsong, H.; Yuan, L. Time series analysis and forecasting of air pollution particulate matter (PM 2.5): An SARIMA and factor analysis approach. *IEEE Access* **2021**, *9*, 41019–41031. [\[CrossRef\]](#)
55. Wang, D.; Ban, X.; Ji, L.; Guan, X.; Liu, K.; Qian, X. An Adaptive Shrinking Grid Search Chaotic Wolf Optimization Algorithm Using Standard Deviation Updating Amount. *Comput. Intell. Neurosci.* **2020**, *2020*, 1–15. [\[CrossRef\]](#)
56. Mahajan, S.; Liu, H.-M.; Tsai, T.-C.; Chen, L.-J. Improving the Accuracy and Efficiency of PM_{2.5} Forecast Service Using Cluster-Based Hybrid Neural Network Model. *IEEE Access* **2018**, *6*, 19193–19204. [\[CrossRef\]](#)
57. Park, C.B.; Sugimoto, N.; Matsui, I.; Shimizu, A.; Tatarov, B.; Kamei, A.; Westphal, D.L. Long-range transport of Saharan dust to east Asia observed with lidars. *Sola* **2005**, *1*, 121–124. [\[CrossRef\]](#)
58. Li, R.; Cui, L.; Meng, Y.; Zhao, Y.; Fu, H. Satellite-based prediction of daily SO₂ exposure across China using a high-quality random forest-spatiotemporal Kriging (RF-STK) model for health risk assessment. *Atmos. Environ.* **2019**, *208*, 10–19. [\[CrossRef\]](#)
59. Huang, K.; Xiao, Q.; Meng, X.; Geng, G.; Wang, Y.; Lyapustin, A.; Gu, D.; Liu, Y. Predicting monthly high-resolution PM_{2.5} concentrations with random forest model in the North China Plain. *Environ. Pollut.* **2018**, *242 Pt A*, 675–683. [\[CrossRef\]](#)
60. Wei, J.; Li, Z.; Lyapustin, A.; Sun, L.; Peng, Y.; Xue, W.; Cribb, M. Reconstructing 1-km-resolution high-quality PM_{2.5} data records from 2000 to 2018 in China: Spatiotemporal variations and policy implications. *Remote Sens. Environ.* **2021**, *252*, 112136. [\[CrossRef\]](#)
61. Wei, J.; Li, Z.; Pinker, R.T.; Wang, J.; Sun, L.; Xue, W.; Li, R.; Cribb, M. Himawari-8-derived diurnal variations in ground-level PM 2.5 pollution across China using the fast space-time Light Gradient Boosting Machine (LightGBM). *Atmos. Chem. Phys.* **2021**, *21*, 7863–7880.
62. Jung, C.R.; Hwang, B.F.; Chen, W.T. Incorporating long-term satellite-based aerosol optical depth, localized land use data, and meteorological variables to estimate ground-level PM_{2.5} concentrations in Taiwan from 2005 to 2015. *Environ. Pollut.* **2018**, *237*, 1000–1010. [\[CrossRef\]](#)

63. Ma, R.; Ban, J.; Wang, Q.; Zhang, Y.; Yang, Y.; He, M.Z.; Li, T. Random forest model based fine scale spatiotemporal O₃ trends in the Beijing-Tianjin-Hebei region in China, 2010 to 2017. *Environ. Pollut.* **2021**, *276*, 116635. [[CrossRef](#)]
64. Hu, X.; Zhang, J.; Xue, W.; Zhou, L.; Che, Y.; Han, T. Estimation of the near-surface ozone concentration with full spatiotemporal coverage across the Beijing-Tianjin-Hebei region based on extreme gradient boosting combined with a WRF-chem model. *Atmosphere* **2022**, *13*, 632. [[CrossRef](#)]
65. Breiman, L. Random forests. *Mach. Learn.* **2001**, *45*, 5–32. [[CrossRef](#)]
66. Chen, T.; Guestrin, C. Xgboost: A scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, San Francisco, CA, USA, 13–17 August 2016; pp. 785–794.
67. Rumelhart, D.E.; Hinton, G.E.; Williams, R.J. Learning representations by back-propagating errors. *Nature* **1986**, *323*, 533–536. [[CrossRef](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.